GREEEN tallgrass prairie restoration: Seeded richness ~ Seeding time * Herbicide pre-treatment (Lux Arbor)

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Contents

Preparation **1**

Statistics **2**
Data exploration . . . 2
Means and deviations . . . 2
Graphs of raw data (Step 2, 6, 7) . . . 4
Outliers, zero-inflation, transformations? (Step 1, 3, 4) . . . 7
Check collinearity part 1 (Step 5) . . . 7
Models . . . 7
Model check . . . 8
DHARMa . . . 8
Check collinearity part 2 (Step 5) . . . 12
Model comparison . . . 12
R2 values . . . 12
AICc . . . 13
Predicted values . . . 13
Summary table . . . 13
Forest plot . . . 14
Effect sizes . . . 15

Session info **18**

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> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(lme4)
library(glmmTMB)
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    herbicide = col_factor(levels = c("0", "1"), ordered = FALSE),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "Lux Arbor",
    richness_type == "seeded_richness",
    treatment_id %in% c("1", "2", "3", "4")
  ) %>%
  mutate(
    treatment = str_c(seeding_time, herbicide, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  mutate(y = richness_1qm + richness_25qm)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##     upper      mean     lower
## 10.492483  9.657343  8.822202
```

```r
median(sites$y)
```

```
## [1] 11
```

```r
sd(sites$y)
```

```
## [1] 5.05199
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##  5% 95%
##   1  17
```

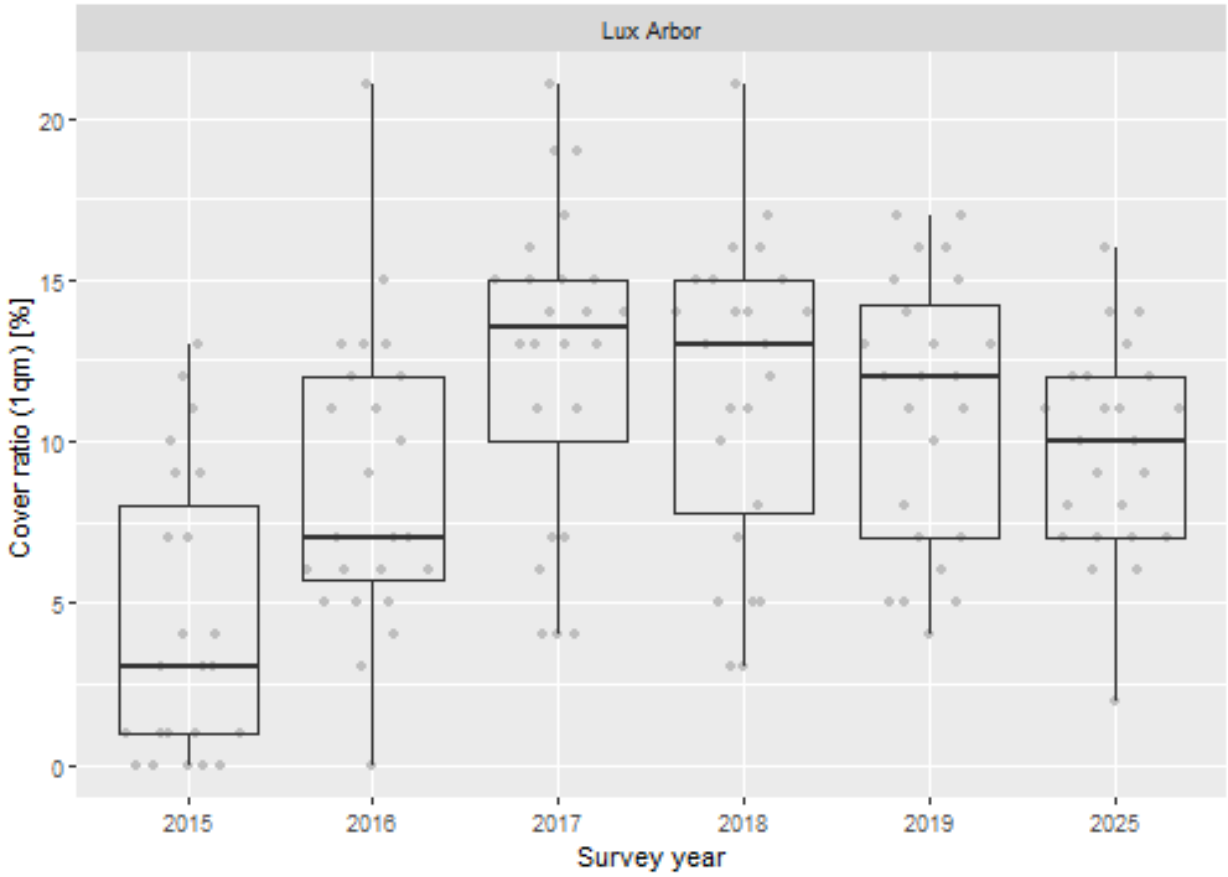
```r
sites %>% count(site, year)
```

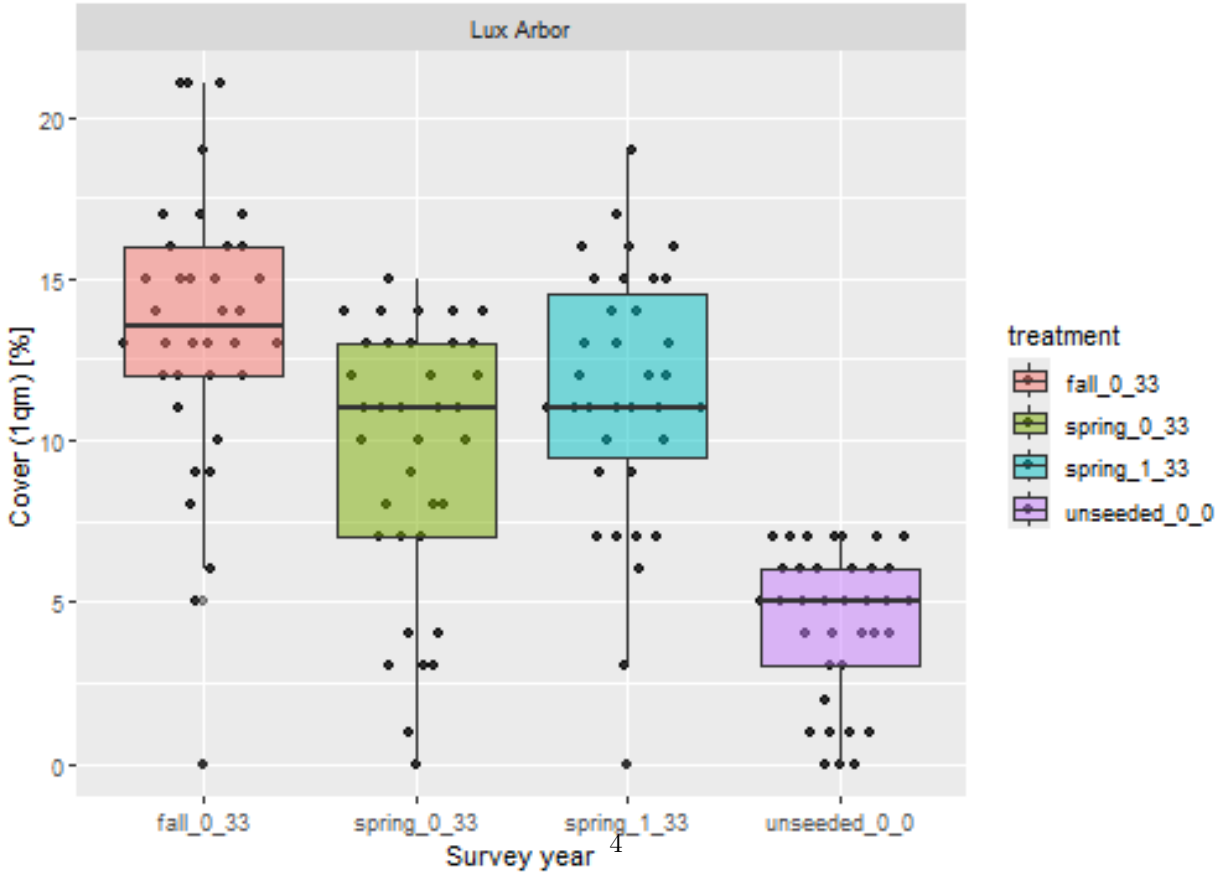
```
## # A tibble: 6 x 3
##   site      year      n
##   <chr>     <fct> <int>
## 1 Lux Arbor 2015     23
## 2 Lux Arbor 2016     24
## 3 Lux Arbor 2017     24
## 4 Lux Arbor 2018     24
## 5 Lux Arbor 2019     24
## 6 Lux Arbor 2025     24
```

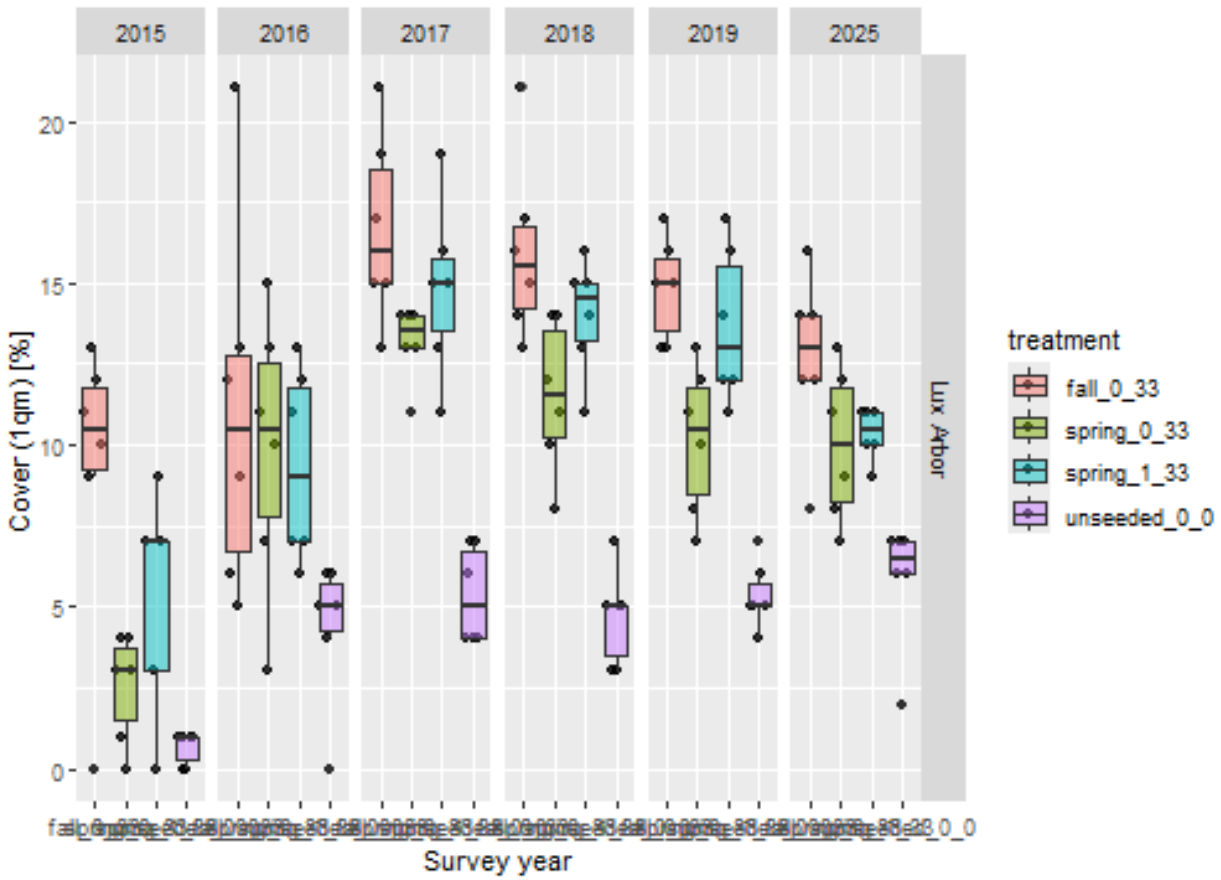
```r
sites %>% count(treatment)
```

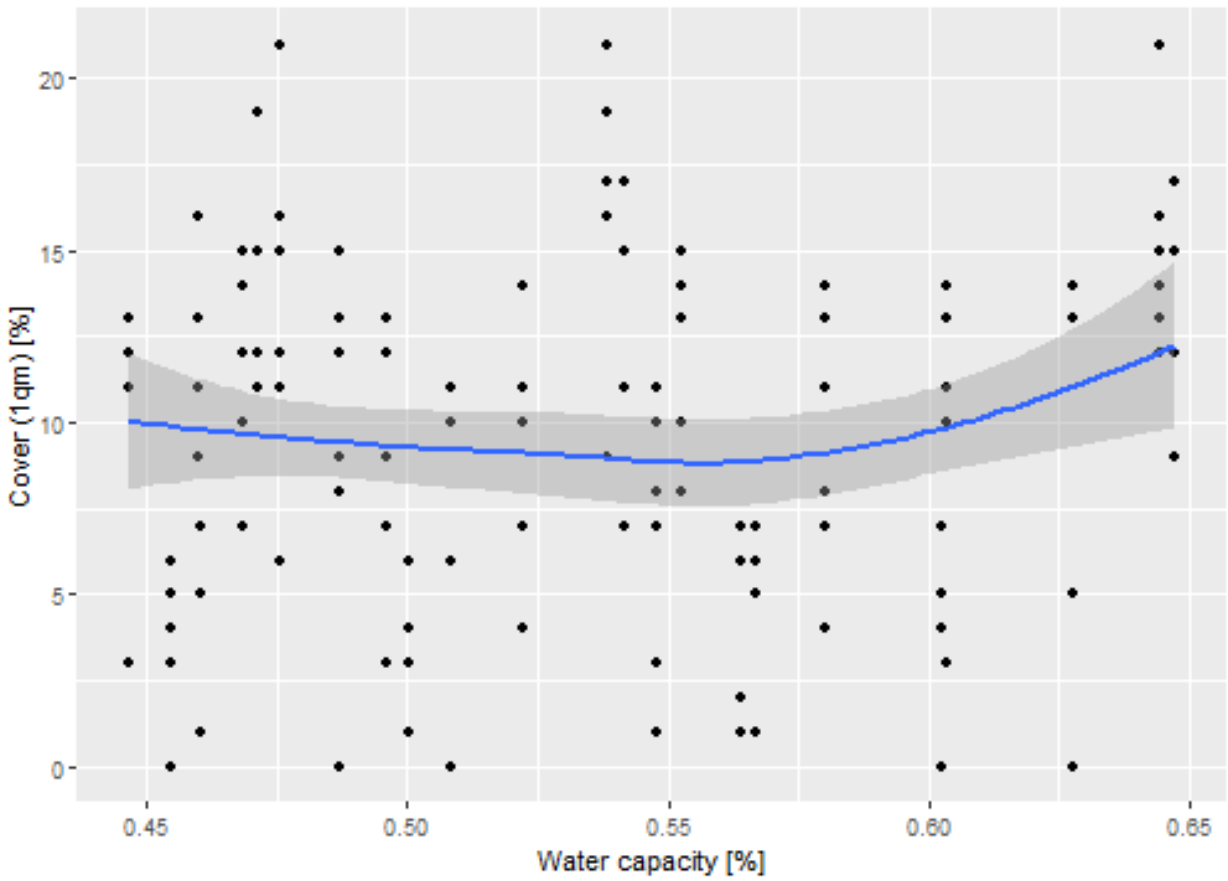
```
## # A tibble: 4 x 2
##   treatment        n
##   <fct>        <int>
## 1 fall_0_33       36
## 2 spring_0_33     36
## 3 spring_1_33     35
## 4 unseeded_0_0    36
```

Graphs of raw data (Step 2, 6, 7)

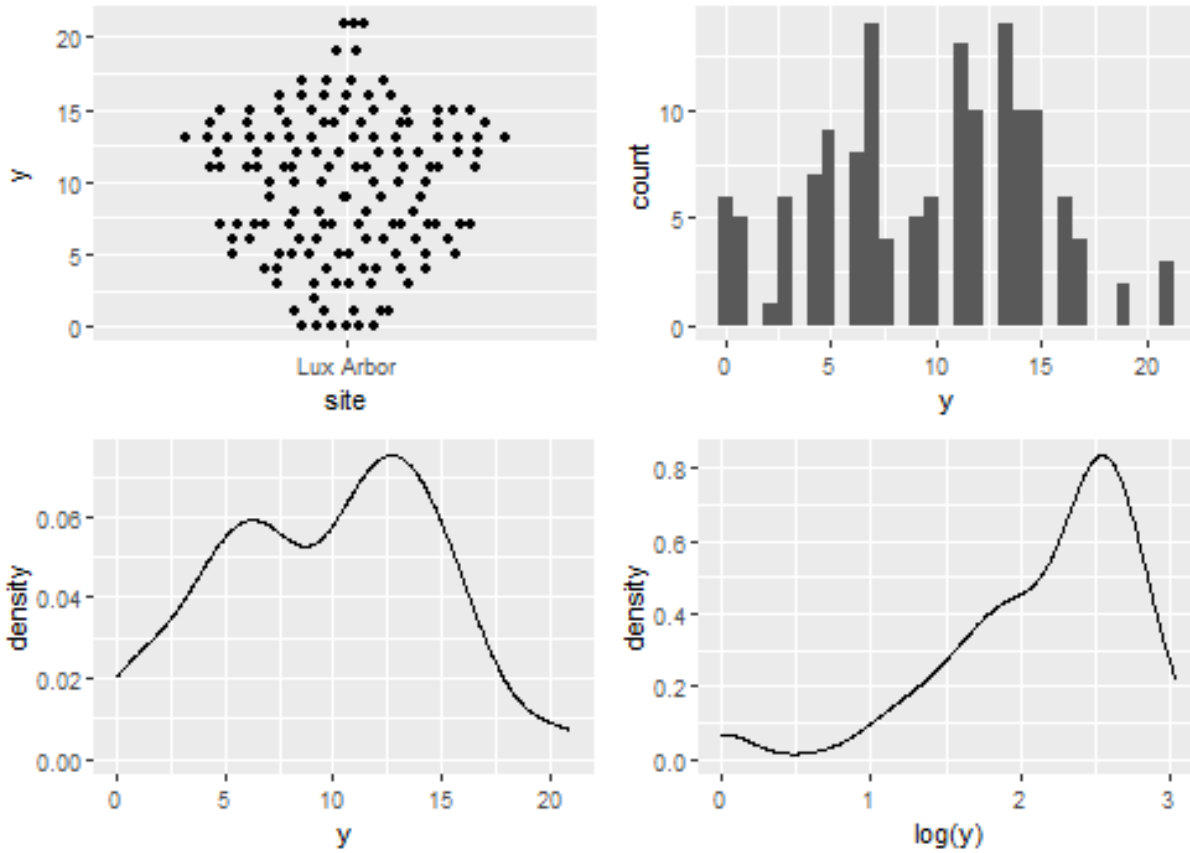





2015
2016
2017
2018
2019
2025
Lux Arbor
Cover (1qm) [%]
0
5
10
15
20
Survey year
treatment
fall_0_33
spring_0_33
spring_1_33
unseeded_0_0


Cover (1qm) [%]
20
15
10
5
0
0.45
0.50
0.55
0.60
0.65
Water capacity [%]

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_richness_timing_herbicide_lux_4.Rdata"))
load(file = here("outputs", "models", "model_richness_timing_herbicide_lux_5.Rdata"))
m_1 <- m4
m_2 <- m5
```

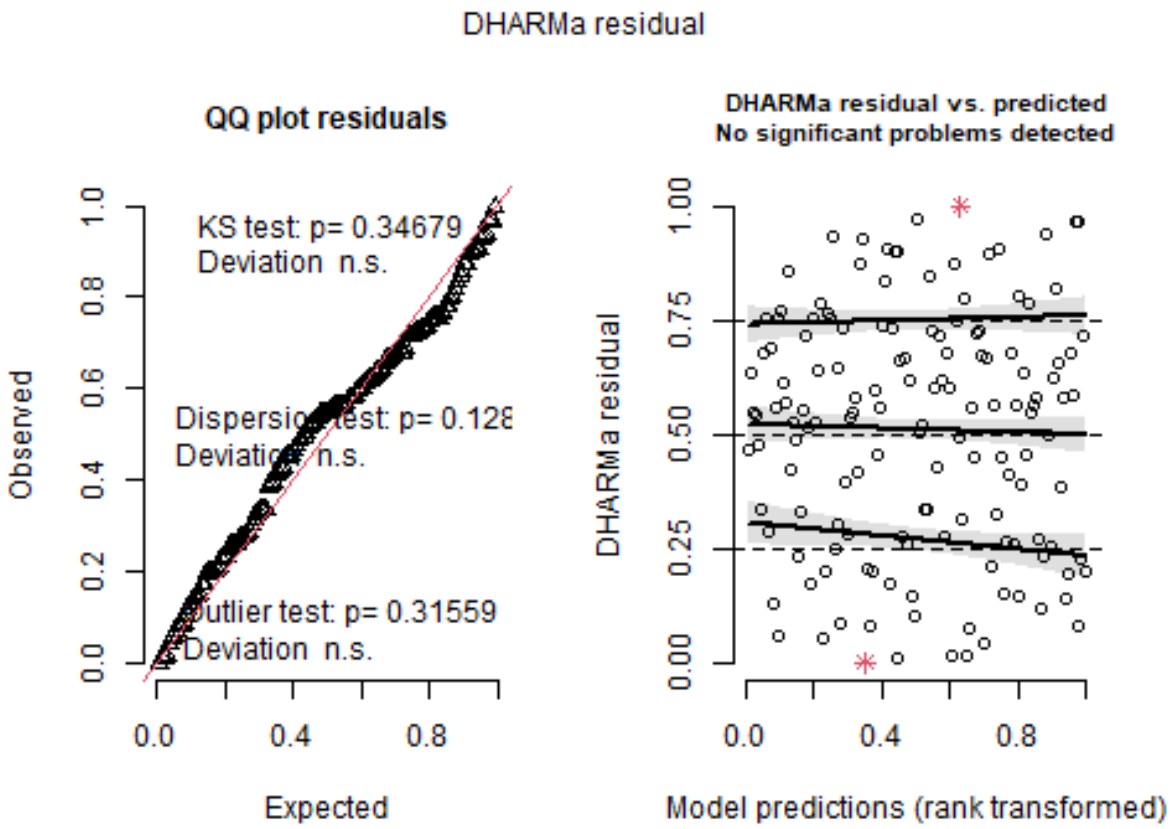
```
formula(m_1)
## y ~ treatment * year + water_cap
formula(m_2)
```

```
## y ~ treatment * (year + water_cap)
```

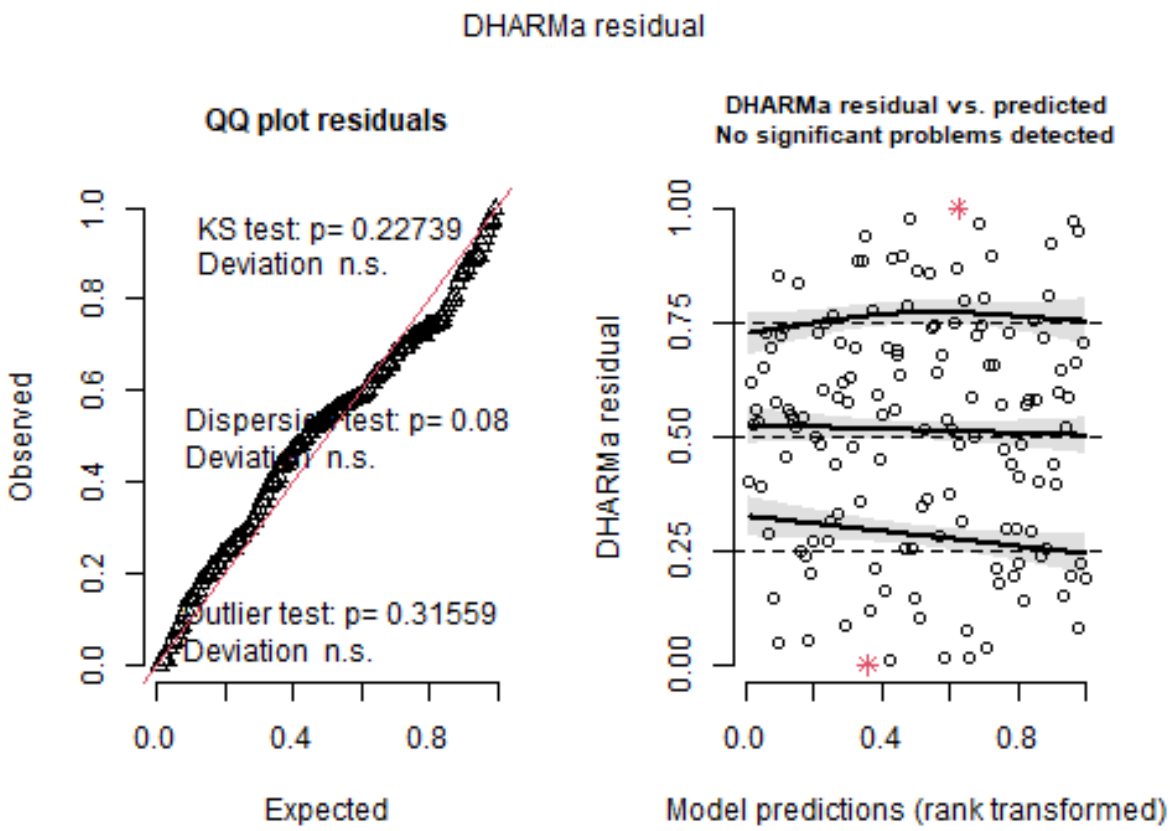
Model check

DHARMa

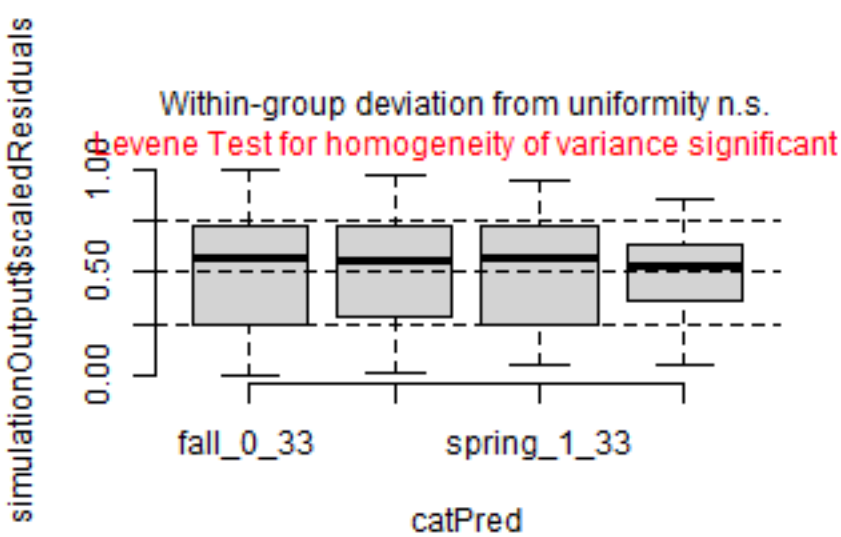
```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



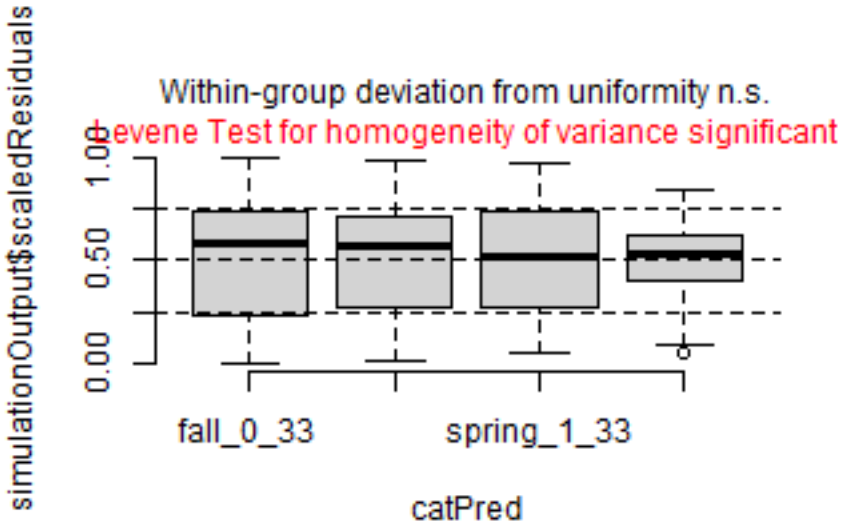

```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



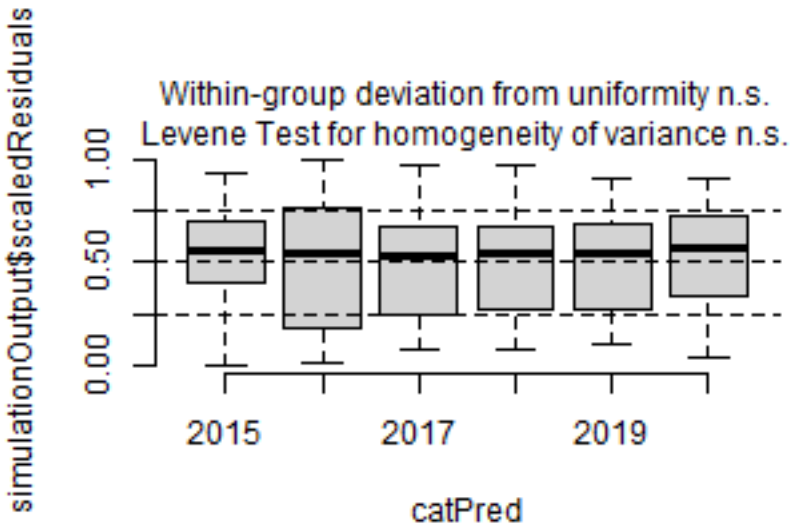

```
plotResiduals(simulation_output_1$scaledResiduals, sites$treatment)
```



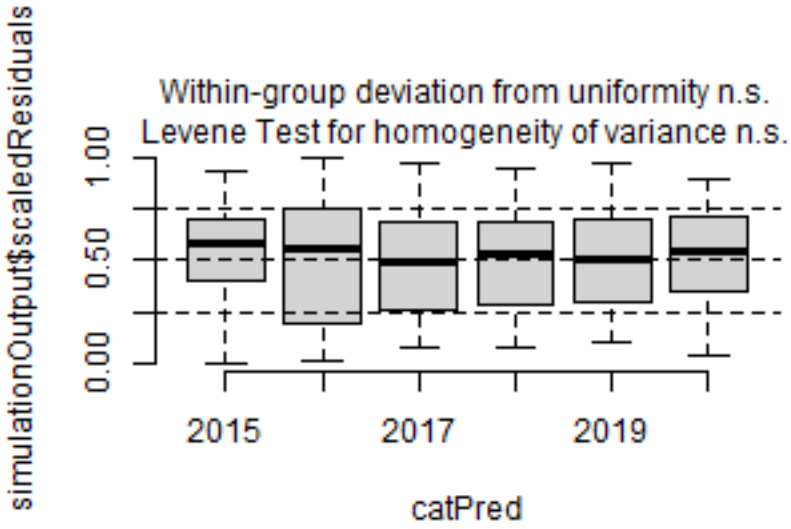

```
plotResiduals(simulation_output_2$scaledResiduals, sites$treatment)
```



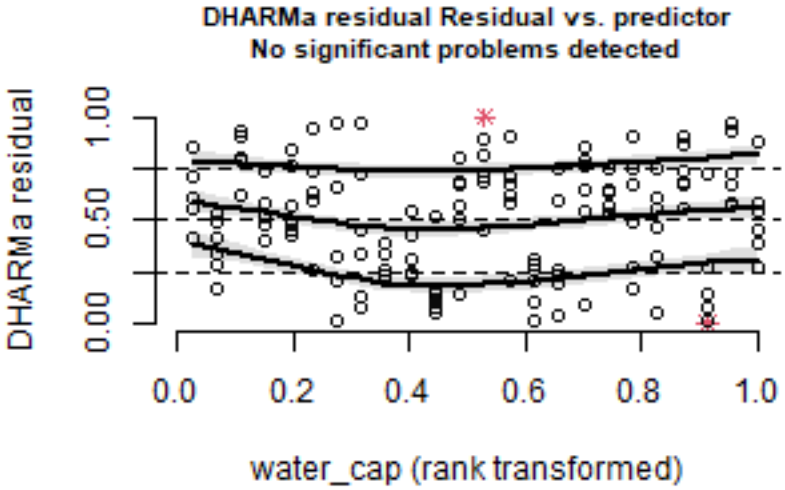

```
plotResiduals(simulation_output_1$scaledResiduals, sites$year)
```



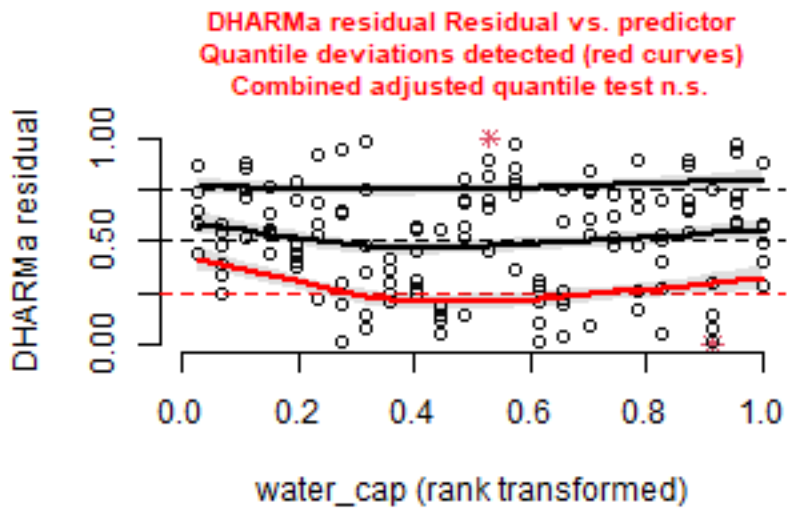

```
plotResiduals(simulation_output_2$scaledResiduals, sites$year)
```




```
plotResiduals(simulation_output_1$scaledResiduals, sites$water_cap)
```




```
plotResiduals(simulation_output_2$scaledResiduals, sites$water_cap)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##                        GVIF Df GVIF^(1/(2*Df))
## treatment        256.494300  3        2.520652
## year             988.195804  5        1.992894
## water_cap          1.371548  1        1.171131
## treatment:year 92236.364539 15        1.463851
```

```
car::vif(m_2)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##                             GVIF Df GVIF^(1/(2*Df))
## treatment           2.364650e+06  3       11.542356
## year                9.881958e+02  5        1.992894
## water_cap           3.673895e+00  1        1.916741
## treatment:year      9.239013e+04 15        1.463932
## treatment:water_cap 1.831351e+06  3       11.061023
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##            R2m       R2c
## [1,] 0.7293559 0.7293559
MuMIn::r.squaredGLMM(m_2)
##            R2m       R2c
## [1,] 0.7295901 0.7295901
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df     AICc
## m_1 26 725.5003
## m_2 29 731.8414
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: y
##                Sum Sq  Df F value    Pr(>F)
## (Intercept)    151.49   1 20.9274 1.186e-05 ***
## treatment      255.95   3 11.7862 8.277e-07 ***
## year           262.22   5  7.2451 5.988e-06 ***
## water_cap       12.81   1  1.7690    0.1861
## treatment:year 160.31  15  1.4764    0.1249
## Residuals      854.16 118
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

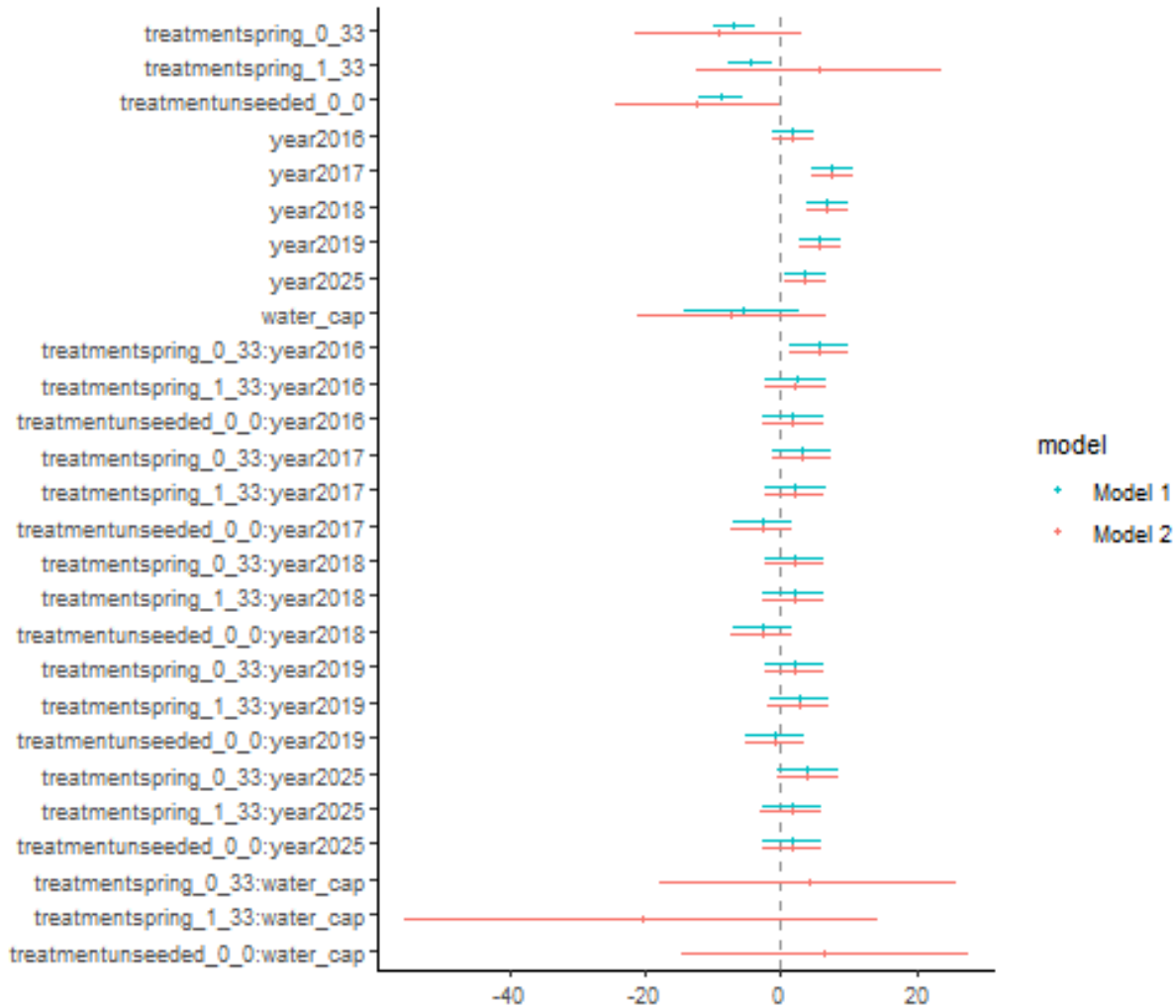
```
summary(m_1)
```

```
##
## Call:
## lm(formula = y ~ treatment * year + water_cap, data = sites)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.8990 -1.5932  0.2658  1.4816  9.7551
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      12.492      2.731   4.575 1.19e-05 ***
## treatmentspring_0_33             -6.952      1.568  -4.433 2.09e-05 ***
## treatmentspring_1_33             -4.459      1.671  -2.669 0.008681 **
## treatmentunseeded_0_0            -8.821      1.572  -5.611 1.35e-07 ***
## year2016                          1.833      1.553   1.180 0.240276
## year2017                          7.500      1.553   4.828 4.17e-06 ***
## year2018                          6.833      1.553   4.399 2.40e-05 ***
## year2019                          5.667      1.553   3.648 0.000395 ***
## year2025                          3.500      1.553   2.253 0.026094 *
## water_cap                        -5.724      4.304  -1.330 0.186072
## treatmentspring_0_33:year2016     5.500      2.197   2.504 0.013657 *
## treatmentspring_1_33:year2016     2.278      2.251   1.012 0.313687
## treatmentunseeded_0_0:year2016    1.833      2.197   0.835 0.405651
## treatmentspring_0_33:year2017     3.167      2.197   1.442 0.152088
## treatmentspring_1_33:year2017     2.111      2.251   0.938 0.350264
```

```
## treatmentunseeded_0_0:year2017   -2.833      2.197  -1.290 0.199651
## treatmentspring_0_33:year2018     2.167      2.197   0.986 0.326004
## treatmentspring_1_33:year2018     1.944      2.251   0.864 0.389473
## treatmentunseeded_0_0:year2018   -2.833      2.197  -1.290 0.199651
## treatmentspring_0_33:year2019     2.000      2.197   0.910 0.364452
## treatmentspring_1_33:year2019     2.778      2.251   1.234 0.219673
## treatmentunseeded_0_0:year2019   -1.000      2.197  -0.455 0.649791
## treatmentspring_0_33:year2025     4.000      2.197   1.821 0.071162 .
## treatmentspring_1_33:year2025     1.611      2.251   0.716 0.475602
## treatmentunseeded_0_0:year2025    1.667      2.197   0.759 0.449550
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.69 on 118 degrees of freedom
## Multiple R-squared:  0.7643, Adjusted R-squared:  0.7164
## F-statistic: 15.94 on 24 and 118 DF,  p-value: < 2.2e-16
```

Forest plot

```r
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

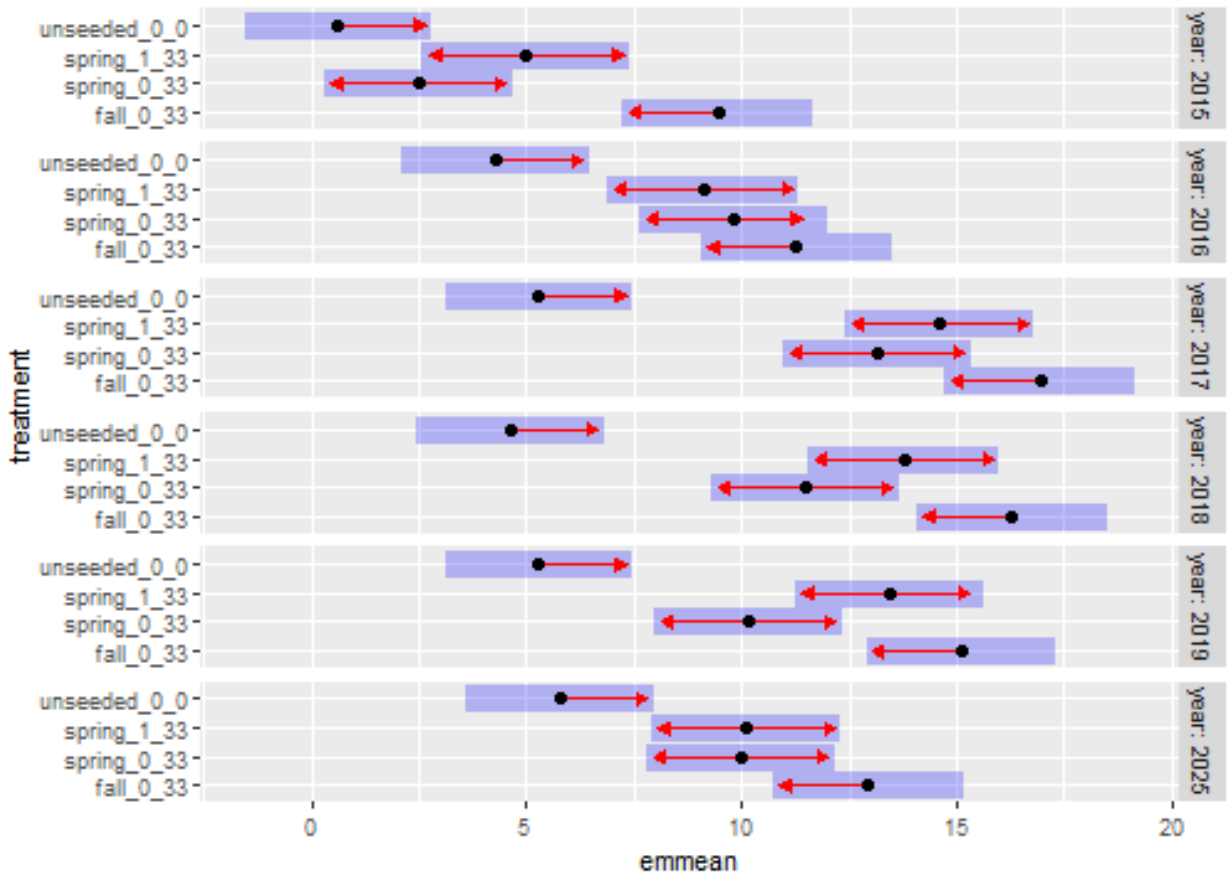
```
(emm <- emmeans(
  m_1,
  revpairwise ~ treatment | year,
  type = "response"
  ))
```

```
## $emmeans
## year = 2015:
##  treatment    emmean   SE  df lower.CL upper.CL
##  fall_0_33     9.444 1.12 118    7.230    11.66
##  spring_0_33   2.493 1.10 118    0.318     4.67
##  spring_1_33   4.985 1.21 118    2.581     7.39
##  unseeded_0_0  0.623 1.10 118   -1.553     2.80
##
## year = 2016:
##  treatment    emmean   SE  df lower.CL upper.CL
```

```
##  fall_0_33    11.278 1.12 118    9.064    13.49
##  spring_0_33   9.826 1.10 118    7.651    12.00
##  spring_1_33   9.096 1.11 118    6.893    11.30
##  unseeded_0_0  4.290 1.10 118    2.114     6.47
##
## year = 2017:
##  treatment    emmean   SE  df lower.CL upper.CL
##  fall_0_33    16.944 1.12 118   14.730    19.16
##  spring_0_33  13.159 1.10 118   10.984    15.33
##  spring_1_33  14.596 1.11 118   12.393    16.80
##  unseeded_0_0  5.290 1.10 118    3.114     7.47
##
## year = 2018:
##  treatment    emmean   SE  df lower.CL upper.CL
##  fall_0_33    16.278 1.12 118   14.064    18.49
##  spring_0_33  11.493 1.10 118    9.318    13.67
##  spring_1_33  13.763 1.11 118   11.560    15.97
##  unseeded_0_0  4.623 1.10 118    2.447     6.80
##
## year = 2019:
##  treatment    emmean   SE  df lower.CL upper.CL
##  fall_0_33    15.111 1.12 118   12.897    17.32
##  spring_0_33  10.159 1.10 118    7.984    12.33
##  spring_1_33  13.430 1.11 118   11.226    15.63
##  unseeded_0_0  5.290 1.10 118    3.114     7.47
##
## year = 2025:
##  treatment    emmean   SE  df lower.CL upper.CL
##  fall_0_33    12.944 1.12 118   10.730    15.16
##  spring_0_33   9.993 1.10 118    7.818    12.17
##  spring_1_33  10.096 1.11 118    7.893    12.30
##  unseeded_0_0  5.790 1.10 118    3.614     7.97
##
## Confidence level used: 0.95
##
## $contrasts
## year = 2015:
##  contrast                   estimate   SE  df t.ratio p.value
##  spring_0_33 - fall_0_33      -6.951 1.57 118  -4.433  0.0001
##  spring_1_33 - fall_0_33      -4.459 1.67 118  -2.669  0.0425
##  spring_1_33 - spring_0_33     2.493 1.64 118   1.523  0.4271
##  unseeded_0_0 - fall_0_33     -8.821 1.57 118  -5.611  <.0001
##  unseeded_0_0 - spring_0_33   -1.869 1.55 118  -1.203  0.6260
##  unseeded_0_0 - spring_1_33   -4.362 1.63 118  -2.669  0.0425
##
## year = 2016:
##  contrast                   estimate   SE  df t.ratio p.value
##  spring_0_33 - fall_0_33      -1.451 1.57 118  -0.926  0.7912
##  spring_1_33 - fall_0_33      -2.181 1.60 118  -1.363  0.5253
##  spring_1_33 - spring_0_33    -0.730 1.56 118  -0.467  0.9661
##  unseeded_0_0 - fall_0_33     -6.987 1.57 118  -4.445  0.0001
##  unseeded_0_0 - spring_0_33   -5.536 1.55 118  -3.563  0.0029
##  unseeded_0_0 - spring_1_33   -4.806 1.56 118  -3.081  0.0135
##
```

```
## year = 2017:
##  contrast                   estimate   SE  df t.ratio p.value
##  spring_0_33 - fall_0_33      -3.785 1.57 118  -2.414  0.0800
##  spring_1_33 - fall_0_33      -2.348 1.60 118  -1.467  0.4608
##  spring_1_33 - spring_0_33     1.437 1.56 118   0.919  0.7945
##  unseeded_0_0 - fall_0_33    -11.654 1.57 118  -7.414  <.0001
##  unseeded_0_0 - spring_0_33   -7.869 1.55 118  -5.065  <.0001
##  unseeded_0_0 - spring_1_33   -9.306 1.56 118  -5.965  <.0001
##
## year = 2018:
##  contrast                   estimate   SE  df t.ratio p.value
##  spring_0_33 - fall_0_33      -4.785 1.57 118  -3.051  0.0147
##  spring_1_33 - fall_0_33      -2.515 1.60 118  -1.571  0.3992
##  spring_1_33 - spring_0_33     2.270 1.56 118   1.453  0.4694
##  unseeded_0_0 - fall_0_33    -11.654 1.57 118  -7.414  <.0001
##  unseeded_0_0 - spring_0_33   -6.869 1.55 118  -4.422  0.0001
##  unseeded_0_0 - spring_1_33   -9.140 1.56 118  -5.858  <.0001
##
## year = 2019:
##  contrast                   estimate   SE  df t.ratio p.value
##  spring_0_33 - fall_0_33      -4.951 1.57 118  -3.158  0.0107
##  spring_1_33 - fall_0_33      -1.681 1.60 118  -1.050  0.7203
##  spring_1_33 - spring_0_33     3.270 1.56 118   2.092  0.1616
##  unseeded_0_0 - fall_0_33     -9.821 1.57 118  -6.247  <.0001
##  unseeded_0_0 - spring_0_33   -4.869 1.55 118  -3.134  0.0115
##  unseeded_0_0 - spring_1_33   -8.140 1.56 118  -5.217  <.0001
##
## year = 2025:
##  contrast                   estimate   SE  df t.ratio p.value
##  spring_0_33 - fall_0_33      -2.951 1.57 118  -1.882  0.2412
##  spring_1_33 - fall_0_33      -2.848 1.60 118  -1.779  0.2886
##  spring_1_33 - spring_0_33     0.104 1.56 118   0.066  0.9999
##  unseeded_0_0 - fall_0_33     -7.154 1.57 118  -4.551  0.0001
##  unseeded_0_0 - spring_0_33   -4.203 1.55 118  -2.705  0.0387
##  unseeded_0_0 - spring_1_33   -4.306 1.56 118  -2.760  0.0334
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```r
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     glmmTMB_1.1.12   lme4_1.1-37
##  [5] Matrix_1.7-3     patchwork_1.3.1  ggbeeswarm_0.7.2 lubridate_1.9.4
##  [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
```

```
## [13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_3.5.2
## [17] tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         compiler_4.5.0
##  [7] mgcv_1.9-3             vctrs_0.6.5            pkgconfig_2.0.3
## [10] crayon_1.5.3           fastmap_1.2.0          backports_1.5.0
## [13] labeling_0.4.3         utf8_1.2.6             ggstance_0.3.7
## [16] promises_1.3.3         rmarkdown_2.29         tzdb_0.5.0
## [19] nloptr_2.2.1           bit_4.6.0              xfun_0.52
## [22] later_1.4.2            parallel_4.5.0         R6_2.6.1
## [25] gap.datasets_0.0.6     stringi_1.8.7          qgam_2.0.0
## [28] RColorBrewer_1.1-3     car_3.1-3              boot_1.3-31
## [31] numDeriv_2016.8-1.1    estimability_1.5.1     Rcpp_1.1.0
## [34] iterators_1.0.14       knitr_1.50             zoo_1.8-14
## [37] parameters_0.27.0      httpuv_1.6.16          splines_4.5.0
## [40] timechange_0.3.0       tidyselect_1.2.1       rstudioapi_0.17.1
## [43] abind_1.4-8            yaml_2.3.10            MuMIn_1.48.11
## [46] doParallel_1.0.17      TMB_1.9.17             codetools_0.2-20
## [49] lattice_0.22-7         plyr_1.8.9             shiny_1.11.1
## [52] withr_3.0.2            bayestestR_0.16.1      coda_0.19-4.1
## [55] evaluate_1.0.4         marginaleffects_0.28.0 pillar_1.11.0
## [58] gap_1.6                carData_3.0-5          foreach_1.5.2
## [61] stats4_4.5.0           reformulas_0.4.1       insight_1.3.1
## [64] generics_0.1.4         vroom_1.6.5            rprojroot_2.0.4
## [67] hms_1.1.3              scales_1.4.0           minqa_1.2.8
## [70] xtable_1.8-4           glue_1.8.0             tools_4.5.0
## [73] data.table_1.17.8      mvtnorm_1.3-3          grid_4.5.0
## [76] rbibutils_2.3          datawizard_1.1.0       nlme_3.1-168
## [79] Rmisc_1.5.1            performance_0.15.0     beeswarm_0.4.0
## [82] vipor_0.4.7            Formula_1.2-5          cli_3.6.5
## [85] gtable_0.3.6           digest_0.6.37          farver_2.1.2
## [88] htmltools_0.5.8.1      lifecycle_1.0.4        mime_0.13
## [91] bit64_4.6.0-1          dotwhisker_0.8.4       MASS_7.3-65
```